BAMS 506 2025W2 BA2
Final Project
Diet Planning Problem

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Chosen Problem: Option 2, Diet Planning Problem

Executive Summary

The Starlight Meal Optimizer was developed to support the “**Feeding Hope**” initiative led by Principal Sally Harper and her team of educators in the small town of Starlight. Following the economic downturn caused by the closure of a major factory, many families in the community have struggled to provide balanced and nutritious meals for their children. This has led to widespread nutritional deficiencies, impacting both physical health and academic performance.

To address this issue, the Starlight Meal Optimizer applies **optimization modeling using Python and Gurobi** to design a **cost effective and nutritionally balanced daily meal plan** for school children. The model minimizes total food costs while meeting essential nutrient requirements established by nutritionists, ensuring that each meal remains diverse and nutritionally complete. Below is the recommended chart for the nutrients:

Daily Nutritional Requirements:

Nutrient	**Min**	**Max**
Calories (kcal)	1800	2400
Fat (g)	60	95
Sodium (mg)	1200	2200
Carbs (g)	240	400
Fiber (g)	30	35
Protein (g)	40	55
Vitamin A (IU)	2000	6000
Vitamin C (mg)	45	1200
Calcium (mg)	1300	3000
Iron (mg)	8	40

Table 1: Nutritional Requirements. “Min” and “Max” denote the minimum and maximum recommended daily intake for a particular nutrient.

Beyond creating a one-day plan, the model extends to a **five-day school-week plan** that ensures variety in daily menus so children do not consume the same foods repeatedly. To ensure variety in the foods, a constraint was introduced that no more than 30% of total daily calories or protein should come from a single food item.

The results of the optimization reveals that nutritious and affordable diets can be achieved even with limited budgets when data-driven decision-making is applied. The solution approach is to maintain the delicate balance between affordability, nutritional value, and variety, and provide a structured approach for the school's five-day meal plan.

Introduction

The town of **Starlight** is facing a serious managerial challenge following the economic collapse caused by the closure of its main factory. This event left many families without stable income, resulting in widespread **food insecurity** and **poor nutrition among children**. After noticing a decline in students' health, concentration, and academic performance, School authorities want to address the issue of inadequate dietary intake. This problem is not just a humanitarian concern but also a resource allocation problem. With the given limited funds how can every child be provided with nutritionally balanced meals each day. The traditional trial and error or intuition-based approach is not sufficient. In order to solve this problem, we need to balance the trade-off between **nutritional adequacy and budget constraints** while ensuring that meals remain diverse and appealing to children, a case of managerial decision-making dilemma. This required finding an efficient, evidence-based method to determine which foods, in what quantities, and at what cost could satisfy all nutritional requirements.

To overcome this challenge, the team turned to **optimization modeling,** a quantitative decision-support tool that systematically evaluates thousands of potential meal combinations to find the most cost-effective and balanced plan. Specifically, a **Linear Programming (LP)** approach was selected and implemented using **Python and Gurobi**. Optimization offers several advantages for this context:

- It **minimizes total food costs** while ensuring all nutrient intake levels stay within recommended bounds.
- It **enforces dietary balance** by limiting the proportion of calories and protein from any single food item.
- It **accommodates managerial flexibility**, allowing the inclusion of additional practical rules such as portion limits or food diversity requirements.

By framing the problem as an optimization model, decision-makers gain a **transparent, data-driven framework** for designing meal plans that are both nutritionally sound and financially sustainable. This approach transforms a complex managerial challenge, constrained budgeting for public nutrition, into a structured analytical problem that yields clear, actionable insights. The solution to LP model will demonstrate how **quantitative analytics and optimization techniques** can empower school administrators and community leaders to make socially impactful, efficient, and sustainable decisions.

Model Formulations

Baseline version (Part A)

The general idea is to minimize the cost of providing food to children while still meeting their baseline nutritional requirements.

Let:

X_i = # of servings of a particular food {i = [1, 2, .., 49]}
p_i = price per serving of a particular food
m_j = variable unit of a particular nutrient {j = [1, 2, … 10]}

Decision Variables:

Per the food_data.xlsx table, let the following decision variables equal
x_1 = # of servings of Almonds
x_2 = # of servings of Apple Raw with skin
…
x_{49} = # of servings of White Tuna in Water

Objective Function:

Min $\sum_{Xi=1}^{49} \sum_{Pi=1}^{49} \{xi * pi\}$ = $p_1*x_1 + p_2*x_2 + … + p_{49}*x_{49} = 0.5x_1 + 0.24x_2 + … + 0.69x_{49}$

Where:
p_1 = price per serving of Almonds
p_2 = price per serving of Apple Raw with skin
…

p_{49} = price per serving of White Tuna in Water

Constraints:

Nutrient Constraints

$\sum_{j=1}^{10} \sum_{i=1}^{49} \{m_j x_i\} \geq m_{jMin}$ and $\sum_{j=1}^{10} \sum_{i=1}^{49} \{m_j x_i\} \leq m_{jMax}$

Written more explicitly:
m_{1Min} $164x_1 + 81.4x_2 + … + 115.6x_3 \geq 1800$
m_{1Max} $164x_1 + 81.4x_2 + … + 115.6x_3 \leq 2400$

$\mathbf{m_{2Min}}$ $14.2x_1 + 0.5x_2 + \ldots + 2.1x_3 \geq 60$
$\mathbf{m_{2Max}}$ $14.2x_1 + 0.5x_2 + \ldots + 2.1x_3 \leq 95$
…
$\mathbf{m_{10Min}}$ $1.1x_1 + 0.2x_2 + \ldots + 0.5x_3 \geq 8$
$\mathbf{m_{10Max}}$ $1.1x_1 + 0.2x_2 + \ldots + 0.5x_3 \leq 40$

Where:
$\mathbf{m_1}$ = calories measured in kcal
$\mathbf{m_2}$ = fat measured in grams
…
$\mathbf{m_{10}}$ = iron measured in milligrams

Balanced Diet Constraints:

Version 1: Per instructions *“No single food item should make up more than 30% of total calories **OR** protein intake”*:

$$x_i * m_1 \leq (0.30) * \sum_{i=1}^{49} (x_i * m_1) + M * (y_i),$$

$$x_i * m_6 \leq (0.30) * \sum_{i=1}^{49} (x_i * m_6) + M * (1 - y_i)$$

Where:
y_i = [0, 1], binary activator
M = “large constant” to make the value purposely high, in excess of any particular possible max nutrient for a given food, for example M = 10,000

Version 2: More realistic version “*No single food time should make up more than 30% of total calories **and** more than 30% of total protein intake*”

$$x_i * m_1 \leq (0.30) * \sum_{i=1}^{49} (x_i * m_1)$$

$$x_i * m_6 \leq (0.30) * \sum_{i=1}^{49} (x_i * m_6)$$

Non-Negativity Constraint:

$\mathbf{x_i} \geq 0$

Model Results

Baseline Model Optimal Solution (Part B)

Food Name	Variable Name	Serving Size	Serving Unit	Optimal Fractional Serving	Absolute Serving Size	Price per Serving	Daily Cost
Banana	x6	1	Fruit	0.179	0.179 Fruit	$0.15	$0.03
Lowfat Milk 2 Percent	x23	1	Cup	2.037	2.037 Cup	$0.23	$0.47
Oranges	x27	1	Fruit	4.698	4.698 Fruit	$0.15	$0.70
Pork	x31	4	Oz	0.565	2.26 Oz	$0.81	$0.46
Potatoes Baked	x32	0.5	Cup	3.149	1.5745 Cup	$0.06	$0.19
Skim Milk	x37	1	Cup	0.971	0.971 Cup	$0.13	$0.13
Spaghetti with Sauce	x38	1.5	Cup	0.614	0.921 Cup	$0.78	$0.48
Total							**$2.45**

Discussions/Recommendations/Conclusions

Reasonable Solution Discussion (Part C)

Based on the optimal solution that was given from our model, we are given a variety of foods that satisfy our constraints. However, the meal plan provided is not reasonable if the sole aim of this optimizer is to provide the nutrition that children need to thrive in school. Based on Canada's food guidelines for parents and children, the Canadian government recommends that for a single day, a child should have a mix of vegetables, fruits, whole grains, and proteins. The current meal plan currently does not contain any vegetables and a child cannot sustain the proper nutrients on just fruits alone and would need a mix of both vegetables and fruits.

To show a more realistic meal plan, the model has been updated with an updated dataset to include the types of food and added constraints so that each food group is represented in our optimized solution. The following has been added to our model:

- Food groups in the "Food_Data.xlsx"
    - Added groups **[Protein, Fruit, Vegetable, Grain/Starch, Dairy]**
- A need to have at least 1 serving from each of the above food groups and having **"Fruit"** and **"Vegetable"** to be between 2-4 servings because they're important in a health diet

Constraints

- Add a constraint for each food group where the total amounts of servings for all foods within the group is greater than 1 (ie. There should be at least 1 serving of each food group). The exception is that "Fruit" and "Vegetable" require servings to both be between 2-4 servings (because we want to have a mix of both).
    - An example would be that all foods under the "Protein" category will have all their servings added up and it should have a total greater or equal to 1

Updated Solution

Food Name	Serving Size	Serving Unit	Optimal Fractional Serving	Absolute Serving Size	Price per Serving	Daily Cost
Apple Raw wSkin	1	Fruit	3.107	3.107 Fruit	$0.24	$0.75
Brown Rice Cooked	0.5	Cup	0.092	0.046 Cup	$0.10	$0.01
Carrots Raw	0.5	Cup (Shredded)	0.131	0.0655 Cup (Shredded)	$0.07	$0.01
Lowfat Milk 2 Percent	1	Cup	2.037	2.037 Cup	$0.23	$0.47
Oranges	1	Fruit	0.893	0.893 Fruit	$0.15	$0.13
Pork	4	Oz	0.562	2.248 Oz	$0.81	$0.46
Potatoes Baked	0.5	Cup	2.701	1.3505 Cup	$0.06	$0.16
Skim Milk	1	Cup	1.566	1.566 Cup	$0.13	$0.20
Spaghetti with Sauce	1.5	Cup	0.557	0.8355 Cup	$0.78	$0.43
White Rice	0.5	Cup	0.351	0.1755 Cup	$0.08	$0.03
Total						**$2.65**

Based on the results of our updated model by adding constraints for food groups, our optimization model has a better variety of foods that contains all the nutrients that a child would need to sustain themselves if they were to eat the same thing everyday while still keeping costs to a minimum. Although the model is still not perfect in creating a realistic solution, it is a step in the right direction. More steps towards a realistic solution would consider the following:

- Adding in constraints for the amount of nutrition that should be consumed throughout the day (ie. Breakfast, Lunch, and Dinner)
- Add constraints where each food needs a minimum serving size to be eaten properly by a child (ie. Cannot have a solution with 0.05 servings of a food such as "baby carrots")
- Have caps on servings per food item (ie. Doesn't make sense to eat 5 oranges a day)
- Include only foods that a child might eat (ie. It's unrealistic to expect a child to eat a raw tomato or just a plain slice of bread)
- Include only finished meals (ie. Remove items such as "Peanut Butter" or "Kale Raw" and instead have have a finished meal that includes these items)

Part D — Five-Day Dietary Plan

Objective and Rationale

Building on the single-day model developed in Parts A-C, we extend the Starlight Meal Optimizer to produce a complete 5-day school-week plan. The primary objective is to maintain daily nutritional compliance while ensuring menu variety, preventing students from being served the same foods repeatedly. This fulfills the project brief to "present a 5-day dietary plan" that is both realistic and appealing.

Model Extension

The single-day model was extended to a weekly horizon with the following structure:

- **Decision Variables**: For each food i and each day d (from 1 to 5), let $x_{i,d}$ be the number of servings of food i on day d.
- **Objective**: Minimize the total weekly cost, calculated as the sum of costs for all servings across all five days: $\sum_{d=1}^{5} \sum_{i} p_i x_i d$, where p_i is the cost per serving.
- **Daily Feasibility Constraints**: For *each* of the five days, the model must satisfy all constraints from the earlier parts:
    - **Nutrient Bounds**: The 10 essential nutrients (Calories, Protein, etc.) must remain within their specified minimum and maximum daily ranges.
    - **Daily Balance**: No single food item may contribute more than 30% of that day's total calories or protein.

    - **Food Groups**: Each day's plan must include at least one serving from the Protein, Dairy, and Grain/Starch groups, and between two and four servings each from the Fruit and Vegetable groups.
- **Weekly Variety Constraints**: To ensure diversity across the week, a new constraint was added:
    - **Limited Repetition**: Any given food item can appear on a maximum of two days in the 5-day plan.

Why could this work for managers?

- **Nutrition Assurance**: The plan guarantees compliance with all 10 nutrient ranges every single day, providing a defensible, guideline-consistent menu.
- **Cost Transparency**: A predictable weekly budget per child supports purchasing and financial planning.
- **Operational Flexibility**: The model can be easily re-run to substitute foods within a group (e.g., swap apples for oranges) to adapt to supply, price changes, or specific allergies without compromising nutrition.

Limited repetition:

We decided to introduce a limited repetition constraint that, on its own, efficiently restricts any item to appear on no more than two days per week. This addresses the variety challenge while still satisfying all original requirements (daily nutrient ranges, 30% per-food caps, and food-group constraints).

For each food $i \in I$:

$$\sum_{d \in D} Yi, d \leq 2$$

Let:

- I: the type of foods
- D = {1,2,3,4,5}: school days
- Yi,d: binary usage:
    - (1) if food i is served on day d
    - (0) otherwise.

As a result, we found no optimal solution to this approach.

```
Optimize a model with 919 rows, 490 columns and 29320 nonzeros
Model fingerprint: 0xaed5ec70
Variable types: 245 continuous, 245 integer (245 binary)
Coefficient statistics:
  Matrix range     [6e-02, 2e+04]
  Objective range  [2e-02, 3e+00]
  Bounds range     [1e+00, 1e+00]
  RHS range        [1e+00, 6e+03]
Presolve added 0 rows and 5 columns
Presolve time: 0.03s
Presolved: 919 rows, 495 columns, 28735 nonzeros
Variable types: 250 continuous, 245 integer (245 binary)

Root relaxation: infeasible, 350 iterations, 0.01 seconds (0.01 work units)

    Nodes    |    Current Node    |     Objective Bounds      |     Work
 Expl Unexpl |  Obj  Depth IntInf | Incumbent    BestBd   Gap | It/Node Time

     0     0 infeasible    0               - infeasible      -     -    0s

Explored 1 nodes (350 simplex iterations) in 0.10 seconds (0.06 work units)
Thread count was 20 (of 20 available processors)

Solution count 0

Model is infeasible
```

The outcome suggest a few potential causes:

With the strict food-group constraints

- We require 2 to 4 servings of both Fruit and Vegetables per day. Over five days, that implies 10 to 20 (Fruit) servings and 10 to 20 (Vegetable) servings in total. At the same time, each individual fruit/vegetable may appear on at most 2 days (2/5 rule). If the menu has too few distinct fruits or vegetables, these requirements can conflict, making it impossible to satisfy the daily minimums across all five days.

Nutrient Constraints

- Some nutrients (e.g., Vitamin A, C, Calcium, Iron) may only be satisfied by a few foods.If those foods are also limited by the 2 day rule, may not be able to meet the daily bounds.

To overcome this issue, we decided to relax the group constraints to a 1 serving daily minimum. This allows the model to maintain daily inclusion of fruits and vegetables while giving enough flexibility to meet tight nutrient caps (e.g., Vitamin A, sodium, fiber) and the 30% per-food limits. It helps to reduce the number of distinct items required under the 2/5 rule, and avoid infeasibility without sacrificing weekly variety.

After running the code in Gurobi, we found that:

```
Optimize a model with 919 rows, 490 columns and 29320 nonzeros
Model fingerprint: 0x3219f604
Variable types: 245 continuous, 245 integer (245 binary)
Coefficient statistics:
  Matrix range     [6e-02, 2e+04]
  Objective range  [2e-02, 3e+00]
  Bounds range     [1e+00, 1e+00]
  RHS range        [1e+00, 6e+03]
Presolve added 0 rows and 5 columns
Presolve time: 0.02s
Presolved: 919 rows, 495 columns, 28735 nonzeros
Variable types: 250 continuous, 245 integer (245 binary)

Root relaxation: infeasible, 292 iterations, 0.01 seconds (0.01 work units)

    Nodes    |    Current Node    |     Objective Bounds      |     Work
 Expl Unexpl |  Obj  Depth IntInf | Incumbent    BestBd   Gap | It/Node Time

     0     0 infeasible    0               - infeasible      -     -    0s

Explored 1 nodes (292 simplex iterations) in 0.07 seconds (0.06 work units)
Thread count was 20 (of 20 available processors)
```

It's infeasible again:

2/5 × daily Fruit/Vegetable minimums (2–4 per day)

- Over 5 days, Fruit ≥ 10 and Veg ≥ 10 servings. If only a small subset of fruits/vegetables are "usable" (because of sodium, vitamin A spikes, fiber minima, cost pressure, etc.), the model runs out of distinct items it can legally repeat under 2/5 and still hit 10 weekly servings while meeting daily min 2. This alone can make the LP relaxation infeasible.

30% "AND" caps (calorie and protein per food per day) × protein minimum

- With a 30% protein cap, we need at least 4 distinct protein contributors per day to reach daily protein minimum (40–55g) if some proteins are constrained by other nutrients/cost. Add 2/5, and we may not have enough usable proteins spread across 5 days.

Vitamin A/C upper bounds × daily Fruit/Vegetable mins

- Carrots, spinach, sweet potato, etc. are vitamin-A heavy. Hitting 2–4 Veg every day while keeping Vit-A ≤ max can become impossible when also under 2/5.

After several trials, we found that adding more weekly constraints made the model increasingly complex and often infeasible. We also tested a weekly 20% cap per item, but the solver still failed to find valid solutions. In addition, we removed the food group constraint, as it consistently pushed the model into the infeasible region as well. Ultimately, we implemented a simpler approach that iterates through each item directly and maintains variety using lightweight logic.

The model will determine how many servings of each food to include per day so that all nutritional targets (calories, protein, vitamins, etc.) stay within healthy limits. It also ensures no single food dominates daily calories or protein (capped at 30% each). To maintain variety, the model links binary on/off indicators to each food's servings and applies a no consecutive day constraint if a food is selected on one day, it's excluded from the next.

Lets:

- Foods $i \in \{1....N\}$, days $d \in \{1,.....,D\}$
- Servings $X_{i,d} \geq 0$
- $y\{i,d\}$, $z\{i,d+1\} \in \{0,1\}$: binary variables for consecutive day pairs
- $\varepsilon > 0$: a small constant (e.g., 0.001)
- M: a large constant (e.g., 1000)

<u>Linking Constraints (Big-M)</u>

$$x\{i,d\} \geq \varepsilon * y\{i,d\} \quad \forall\ i,\ d = 1,\dots,D-1$$
$$x\{i,d\} \leq M * y\{i,d\} \quad \forall\ i,\ d = 1,\dots,D-1$$
$$x\{i,d+1\} \geq \varepsilon * z\{i,d+1\} \quad \forall\ i,\ d = 1,\dots,D-1$$
$$x\{i,d+1\} \leq M * z\{i,d+1\} \quad \forall\ i,\ d = 1,\dots,D-1$$

<u>No-Consecutive-Day Rule</u>

$$y\{i,d\} + z\{i,d+1\} \leq 1 \quad \forall\ i,\ d = 1,\dots,D-1$$

The tables below provide the detailed cost breakdown for the final 5-day meal plan. The cost for each day is calculated from the sum of the costs of the individual food items selected by the optimizer. The sum of these daily costs forms the total weekly budget.

(Note: Item costs are rounded to the nearest cent. Values shown as $0.00 represent costs less than $0.01.)

Monday (Day 1)

Food Item	Optimal Servings	Item Cost
Almonds	1.029 servings	$0.51
Apple Raw wSkin	2.577 servings	$0.62
Banana	5.148 servings	$0.77
Cheddar Cheese	2.357 servings	$0.82
Lowfat Milk 2 Percent	2.037 servings	$0.47
Mixed Nuts	0.303 servings	$0.18
Potatoes Baked	0.357 servings	$0.02
Spaghetti With Sauce	0.416 servings	$0.32
Spinach Raw	0.669 servings	$0.17
White Rice	0.967 servings	$0.08

Monday Total		**$4.16**

Tuesday (Day 2)

Food Item	Optimal Servings	Item Cost
Grapes	8.518 servings	$2.73
Oranges	8.170 servings	$1.23
Pork	0.760 servings	$0.62
White Bread	6.886 servings	$0.41
Skim Milk	1.963 servings	$0.26
Kale Raw	0.253 servings	$0.10
Brown Rice Cooked	0.037 servings	$0.00
Tofu	0.005 servings	$0.00
Tuesday Total		**$5.35**

Wednesday (Day 3)

Food Item	Optimal Servings	Item Cost
Almonds	1.029 servings	$0.51
Apple Raw wSkin	2.577 servings	$0.62
Banana	5.148 servings	$0.77
Cheddar Cheese	2.357 servings	$0.82
Lowfat Milk 2 Percent	2.037 servings	$0.47
Mixed Nuts	0.303 servings	$0.18
Potatoes Baked	0.357 servings	$0.02
Spaghetti With Sauce	0.416 servings	$0.32
Spinach Raw	0.669 servings	$0.17
White Rice	0.967 servings	$0.08
Wednesday Total		**$4.16**

Thursday (Day 4)

Food Item	Optimal Servings	Item Cost
Grapes	8.518 servings	$2.73
Oranges	8.170 servings	$1.23
Pork	0.760 servings	$0.62
White Bread	6.886 servings	$0.41
Skim Milk	1.963 servings	$0.26
Kale Raw	0.253 servings	$0.10
Brown Rice Cooked	0.037 servings	$0.00
Tofu	0.005 servings	$0.00
Thursday Total		**$5.35**

Friday (Day 5)

Food Item	Optimal Servings	Item Cost

Almonds	1.029 servings	$0.51
Apple Raw wSkin	2.577 servings	$0.62
Banana	5.148 servings	$0.77
Cheddar Cheese	2.357 servings	$0.82
Lowfat Milk 2 Percent	2.037 servings	$0.47
Mixed Nuts	0.303 servings	$0.18
Potatoes Baked	0.357 servings	$0.02
Spaghetti With Sauce	0.416 servings	$0.32
Spinach Raw	0.669 servings	$0.17
White Rice	0.967 servings	$0.08
Friday Total		**$4.16**

Total Weekly Cost Calculation

The total weekly budget is the sum of the daily costs:

$4.16 (Mon) + $5.35 (Tue) + $4.16 (Wed) + $5.35 (Thu) + $4.16 (Fri) = **$23.18 per week**

We tried several weekly rules, but they made the model complex and often infeasible. So we chose a simpler, sturdier setup: keep the **daily 30% cap** on any one food's calories/protein, add a **no-consecutive-day** rule for variety, and let the optimizer meet all **daily nutrient targets**. This approach is easy to explain, runs reliably, and gives clear itemized costs by day. The final 5-day plan totals **$23.18**, reflecting a practical balance between variety, nutrition, and cost. It's also easier to maintain for real school operation as menus can be tweaked without breaking the model. If needed, we can gradually add gentle improvements (e.g., soft nudges for food groups or light penalties for repeats) while keeping the system transparent and feasible.

Conclusion

The Starlight Meal Optimizer demonstrates that a linear programming approach can solve a complex, emotionally sensitive problem. The team was able to provide a reasonable solution to feed the children nutritiously on a tight budget. Starting from a one-day formulation and extending to a five-day plan, the model consistently meets essential nutrient ranges while capping the share of daily calories and protein from any single food to preserve balance and avoid over-reliance. Where more substantial "variety" rules created infeasibility under real-world nutrient bounds, we adopted a pragmatic design: daily 30% caps plus a no-consecutive-day rule. The result is a transparent weekly menu that remains feasible, nutritionally adequate, and easy to operate, with an illustrative weekly cost of $23.18 per child.
Managerially, this delivers three immediate wins.

To start with, the team delivered nutrition assurance, and the suggested day-by-day menus meet the recommended ranges for the ten key nutrients. Secondly, the solution is optimized to minimize costs, thus making the plan cost-effective. The plan also supports a weekly budget, enabling informed purchasing, effective inventory management, and successful vendor negotiations. Finally, the team developed an operational model that can be quickly re-run to reflect price changes, supply disruptions, or substitutions within food groups (e.g., swapping apples for oranges) without sacrificing compliance. At the same time, the work surfaces important limitations. Strict variety constraints (e.g., "2-of-5 days" per item) can clash with tight nutrient ceilings (notably Vitamin A and sodium) and with minimum serving requirements for fruits/vegetables, especially when only a few items adequately meet those nutrient needs. Fractional servings, although standard in optimization, may be operationally awkward for kitchen teams, and palatability or cultural preferences are not yet factored in. Finally, the current design is deterministic, prices and nutrient profiles are treated as fixed, and allergy/dietary restrictions are handled outside the core model.

The team recommends enhancements in the future iteration for addressing the limitations of the model:

1. Relaxing complex variety rules with temperate penalties for repetition to retain feasibility while nudging diversity

2. Introduce an integer constraint or minimum serving granularity on items based on kitchen realism.
3. Introducing soft budget scenarios and sensitivity analysis to plan for price swings
4. Piloting a feedback loop simple taste survey to learn and adapt preferences over time.

Overall, the project provides a usable and auditable foundation that a school can adopt immediately and refine iteratively. The team was able to demonstrate that even with inflexible resource availability, a Linear Programming-based solution can certainly promote children's health, stabilize costs, and keep the operations manageable, transforming a challenging civic issue into a sustainable, community-focused solution.